(i) There are special sets of numbers a, b and c that makes the Pythagorean relationship true and these sets of special numbers are called Pythagorean triplets. Example: (3, 4, 5) is a Pythagorean triplet.

(ii) Let **k** be any positive integer greater than 1 and (a, b, c) be a Pythagorean triplet, then (ka, kb, kc) is also a Pythagorean triplet.

Examples:

k	**(3,4,5)**	**(5,12,13)**
2k	(6,8,10)	(10,24,26)
3k	(9,12,15)	(15,36,39)
4k	(12,16,20)	(20,48,52)

So, it is clear that we can have infinitely many Pythagorean triplets just by multiplying any Pythagorean triplet by **k.**

We shall now see a few examples on the use of Pythagoras theorem in problems.

Example 5.11

In the figure, $AB \perp AC$

a) What type of $\varnothing$ is ABC?

b) What are AB and AC of the ΔABC ?

c) What is CB called as ?

d) If $AC = AB$ then, what is the measure of $\angle B$ and $\angle C$?

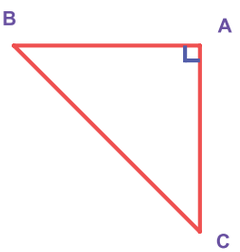


Fig. 5.25

Solution:

a) ΔABC is right angled as $AB \perp AC$ at A.

b) AB and AC are legs of ΔABC .

c) CB is called as the hypotenuse.

d) $\angle B + \angle C = 90^\circ$ and equal angles are opposite to equal sides. Hence, $\angle B = \angle C = \frac{90^\circ}{2} = 45^\circ$

Example 5.12

Can a right triangle have sides that measure 5*cm*, 12*cm* and 13*cm*?

Solution:

Take $a = 5$, $b = 12$ and $c = 13$

Now, $a^2 + b^2 = 5^2 + 12^2 = 25 + 144 = 169 = 13^2 = c^2$

By the converse of Pythagoras theorem, the triangle with given measures is a right angled triangle.

Example 5.13

A 20-feet ladder leans against a wall at height of 16 feet from the ground. How far is the base of the ladder from the wall?

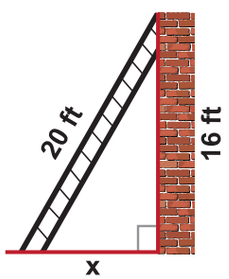


Fig. 5.26

Solution:

The ladder, wall and the ground form a right triangle with the ladder as the hypotenuse. From the figure, by Pythagoras theorem,

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